Top 15 Java Programming Interview Questions with Answers

1. Reverse a String (Without using inbuilt methods)

```
public class ReverseString {
    public static void main(String[] args) {
        String str = "hello";
        String reversed = "";
        for(int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }
        System.out.println("Reversed: " + reversed);
    }
}
```

2. Check if a String is a Palindrome

```
public class PalindromeCheck {
    public static void main(String[] args) {
        String str = "madam";
        String reversed = "";
        for(int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }
        if(str.equals(reversed)) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not a palindrome");
        }
    }
}
```

3. Swap Two Numbers Without Using a Third Variable

```
public class SwapNumbers {
    public static void main(String[] args) {
        int a = 5, b = 10;
        a = a + b;
        b = a - b;
        a = a - b;
        System.out.println("a: " + a + ", b: " + b);
    }
}
```

4. Frequency of Characters in a String

```
import java.util.HashMap;
```

```java
public class CharFrequency {
    public static void main(String[] args) {
        String str = "hello";
        HashMap<Character, Integer> map = new HashMap<>();
        for(char c : str.toCharArray()) {
            map.put(c, map.getOrDefault(c, 0) + 1);
        }
        System.out.println(map);
    }
}
```

5. Remove Duplicate Characters from a String

```java
public class RemoveDuplicates {
    public static void main(String[] args) {
        String str = "programming";
        String result = "";
        for(int i = 0; i < str.length(); i++) {
            if(!result.contains(String.valueOf(str.charAt(i)))) {
                result += str.charAt(i);
            }
        }
        System.out.println(result);
    }
}
```

6. Print Fibonacci Series (Without Recursion)

```java
public class Fibonacci {
    public static void main(String[] args) {
        int n = 10, a = 0, b = 1;
        for(int i = 0; i < n; i++) {
            System.out.print(a + " ");
            int sum = a + b;
            a = b;
            b = sum;
        }
    }
}
```

7. Factorial of a Number (Iterative)

```java
public class Factorial {
    public static void main(String[] args) {
        int num = 5, fact = 1;
        for(int i = 1; i <= num; i++) {
            fact *= i;
        }
        System.out.println("Factorial: " + fact);
    }
}
```

8. Check if a Number is Prime

```
public class PrimeCheck {
    public static void main(String[] args) {
        int num = 7, count = 0;
        for(int i = 2; i < num; i++) {
            if(num % i == 0) count++;
        }
        System.out.println(count == 0 ? "Prime" : "Not Prime");
    }
}
```

9. Find Second Largest in Array

```
public class SecondLargest {
    public static void main(String[] args) {
        int[] arr = {5, 3, 9, 1, 6};
        int first = Integer.MIN_VALUE, second = Integer.MIN_VALUE;
        for(int num : arr) {
            if(num > first) {
                second = first;
                first = num;
            } else if(num > second && num != first) {
                second = num;
            }
        }
        System.out.println("Second Largest: " + second);
    }
}
```

10. Sort Array Without Arrays.sort()

```
public class SortArray {
    public static void main(String[] args) {
        int[] arr = {5, 3, 8, 1};
        for(int i = 0; i < arr.length; i++) {
            for(int j = i+1; j < arr.length; j++) {
                if(arr[i] > arr[j]) {
                    int temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
        for(int i : arr) System.out.print(i + " ");
    }
}
```

11. Armstrong Number

```
public class Armstrong {
    public static void main(String[] args) {
```

```
        int num = 153, temp = num, sum = 0;
        while(temp > 0) {
            int digit = temp % 10;
            sum += digit * digit * digit;
            temp /= 10;
        }
        System.out.println(num == sum ? "Armstrong" : "Not Armstrong");
    }
}
```

12. Count Vowels and Consonants

```
public class VowelConsonantCount {
    public static void main(String[] args) {
        String str = "hello world";
        int vowels = 0, consonants = 0;
        str = str.toLowerCase();
        for(char c : str.toCharArray()) {
            if(c >= 'a' && c <= 'z') {
                if("aeiou".indexOf(c) != -1)
                    vowels++;
                else
                    consonants++;
            }
        }
        System.out.println("Vowels: " + vowels + ", Consonants: " +
consonants);
    }
}
```

13. Find Missing Number in Array

```
public class MissingNumber {
    public static void main(String[] args) {
        int[] arr = {1, 2, 4, 5};
        int n = 5;
        int total = n * (n + 1) / 2;
        int sum = 0;
        for(int num : arr) sum += num;
        System.out.println("Missing Number: " + (total - sum));
    }
}
```

14. Reverse an Integer Without String

```
public class ReverseInteger {
    public static void main(String[] args) {
        int num = 1234, rev = 0;
        while(num != 0) {
            rev = rev * 10 + num % 10;
            num /= 10;
        }
```

```
        System.out.println("Reversed: " + rev);
    }
}
```

15. Print Star Patterns

```
public class StarPattern {
    public static void main(String[] args) {
        for(int i = 1; i <= 5; i++) {
            for(int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```